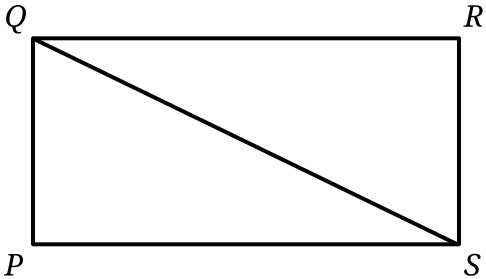


Question ID: 2c3aefc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

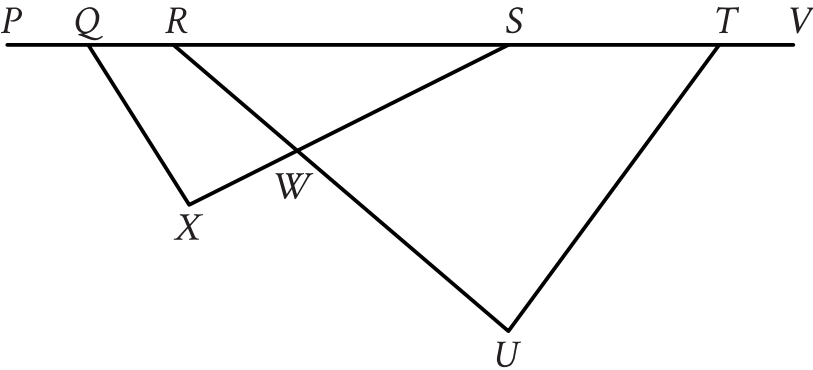


Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



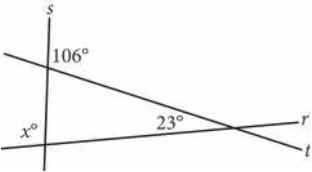
Note: Figure not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment PV , and line segment RU intersects line segment SX at point W . The measure of $\angle SQX$ is 48° , the measure of $\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

Question ID: f88f27e5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

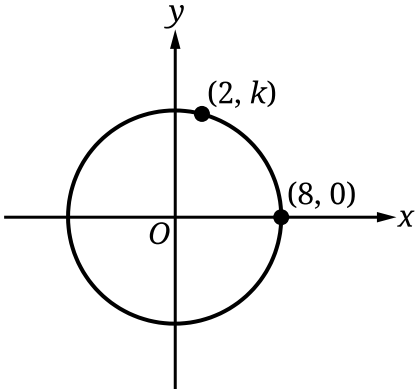
Question
Intersecting lines r, s, and t are shown below.



What is the value of x ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Note: Figure not drawn to scale.

The circle shown has its center at $(0, 0)$. What is the value of k ?

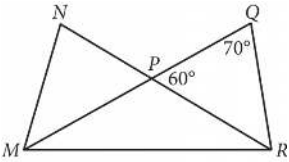
Answer

- A. 6
- B. 7
- C. 8
- D. $\sqrt{60}$

Question ID: 947a3cde

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

Question ID: deff8a2f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

Answer

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

Question ID: 1fe41c6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n ?

Answer

- A. 2,100
- B. 1,680
- C. 840
- D. 420

Question ID: 27d075d8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question
For two acute angles, $\angle Q$ and $\angle R$, $\cos (Q)=\sin (R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x+52$ and $4x+13$, respectively. What is the value of x ?

Answer

- A. 5
- B. 13
- C. 23
- D. 38

Question ID: 1dc7e423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

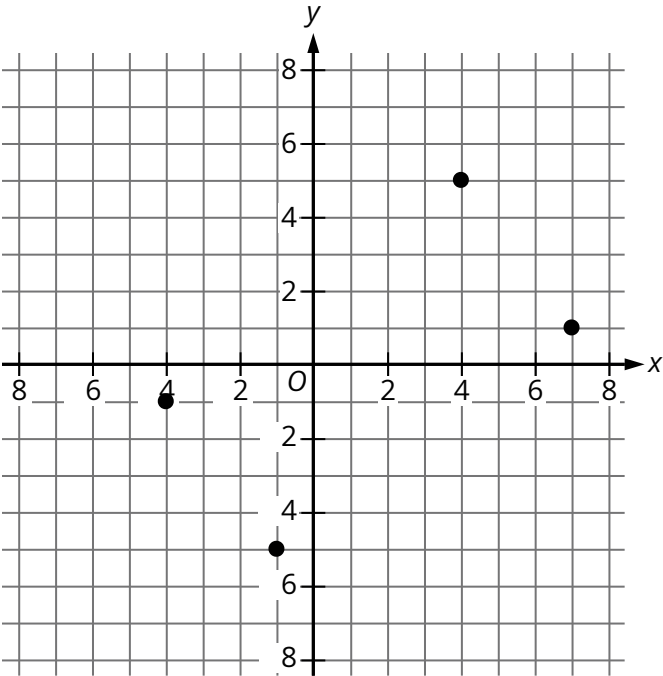
In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

Answer

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Question ID: 8493fd10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

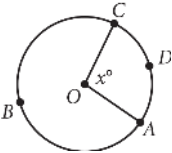
The given equation, where t is a constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{83}$. What is the value of t ?

Answer

- A. 73
- B. 71
- C. 37
- D. 35

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



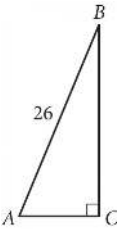
The circle above has center O, the length of arc $\widehat{ADC}$ is 5π , and $x = 100$. What is the length of arc $\widehat{ABC}$?

Answer

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

Question ID: f7dbde16

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

Answer

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

Question ID: 58c26db8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

Answer

- A. 9
- B. $9\sqrt{2}$
- C. 18
- D. $18\sqrt{2}$

Question ID: 35d37640

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

Answer

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π

Question ID: 2266984b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$x^2 + 20x + y^2 + 16y = -20$

The equation above defines a circle in the xy-plane. What are the coordinates of the center of the circle?

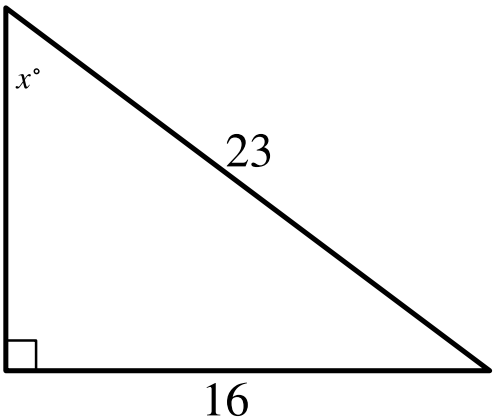
Answer

- A. $(-20, -16)$
- B. $(-10, -8)$
- C. $(10, 8)$
- D. $(20, 16)$

Question ID: 1429dcdf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



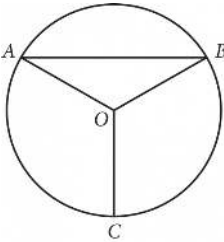
Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^\circ$?

Question ID: 69b0d79d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc $\widehat{AB}$?

Answer

- A. 9π
- B. 12π
- C. 15π
- D. 18π

Question ID: 502d9690

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

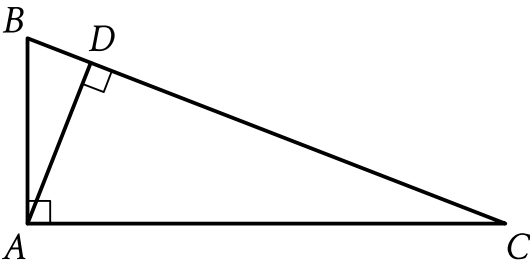
Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is **648** square inches, and the area of rectangle $EFGH$ is **72** square inches. The length of the longest side of rectangle $ABCD$ is **36** inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

Answer

- A. **4**
- B. **9**
- C. **12**
- D. **36**

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



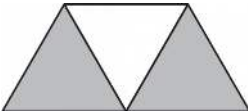
Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . If the length of $\overline{BD}$ is 5 and the tangent of angle CAD is 1.3, what is the length of $\overline{AD}$?

Question ID: 4c95c7d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

Answer

- A. $2\sqrt{3}$
- B. $4\sqrt{3}$
- C. $8\sqrt{3}$
- D. 16

Question ID: b8a225ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

Question ID: f389569d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

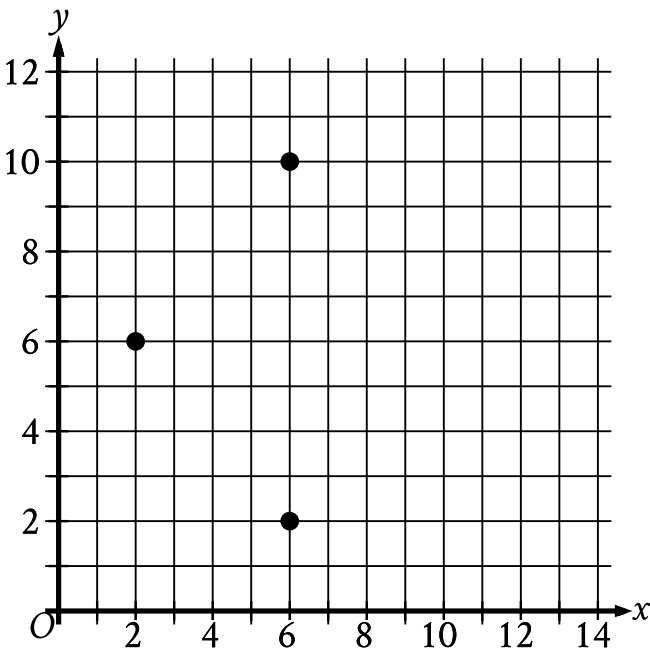
What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

Answer

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

Question ID: e3158182

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t ?

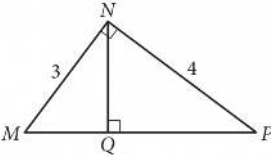
Answer

- A. $6t + \pi$
- B. $6t$
- C. $5t + \pi$
- D. $5t + \frac{\pi}{2}$

Question ID: 740bf79f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, what is the length of $\overline{NQ}$?

Answer

- A. 2.2
- B. 2.3
- C. 2.4
- D. 2.5

Question ID: ab176ad6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The equation $(x + 6)^2 + (y + 3)^2 = 121$ defines a circle in the xy-plane. What is the radius of the circle?

Question ID: 179edb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **18 cm**. The volume of this cone is **$n\pi \text{ cm}^3$** . What is the value of **n** ?

Question ID: 3e577e4a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

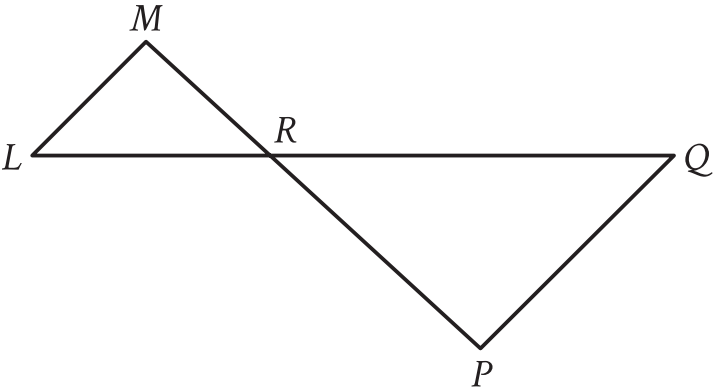
A circle in the xy -plane has its center at $(-4, -6)$. Line k is tangent to this circle at the point $(-7, -7)$. What is the slope of line k ?

Answer

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

Answer

- A. $\frac{576}{49}$
- B. $\frac{144}{7}$
- C. 63
- D. $\frac{441}{4}$

Question ID: b0dc920d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

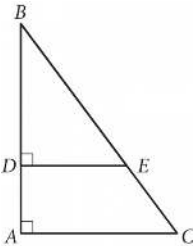
A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V , in cubic inches, of such containers, where r is the radius, in inches?

Answer

- A. $V = 4\pi^3$
- B. $V = \pi(2r)^3$
- C. $V = \pi^2 + 4\pi$
- D. $V = \pi^3 + 4\pi^2$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

Question ID: fecc446d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

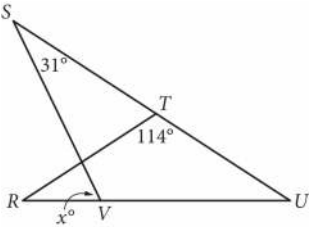
Answer

- A. $-14x + 1,540$
- B. $14x - 320$
- C. $-28x + 1,720$
- D. 360

Question ID: bd7f6e30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, $RT = TU$. What is the value of x ?

Answer

- A. 72
- B. 66
- C. 64
- D. 58

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is **2** times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is **5** square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

Answer

- A. **7**
- B. **10**
- C. **20**
- D. **25**

Question ID: 9e44284b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, the graph of $2x^2 - 6x + 2y^2 + 2y = 45$ is a circle. What is the radius of the circle?

Answer

- A. 5
- B. 6.5
- C. $\sqrt{40}$
- D. $\sqrt{50}$

Question ID: 98e02472

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle XYZ , angles X and Z are acute angles. Point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. The measure of $\angle XZY$ is a° , the measure of $\angle XYR$ is b° , and the length of $\overline{XY}$ is 19. If $\sin a^\circ = \frac{12}{13}$, what is the value of $\tan b^\circ$?

Question ID: e6c557f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

Answer

- A. $\frac{9}{7,744}$
- B. $\frac{27}{3,872}$
- C. $\frac{3}{88}$
- D. $\frac{9}{44}$

Question ID: 568d66a7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

Answer

- A. 47
- B. $47\sqrt{2}$
- C. 94
- D. $94\sqrt{2}$

Question ID: 322a6dfe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question
Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R correspond to W , X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

- Answer**
- A. 5°
 - B. 27°
 - C. 45°
 - D. 135°

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

$RS = 440$

$ST = 384$

$TR = 584$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

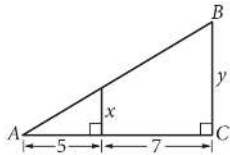
Answer

- A. $\frac{48}{73}$
- B. $\frac{55}{73}$
- C. $\frac{48}{55}$
- D. $\frac{55}{48}$

Question ID: eeb4143c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

Question ID: 5b2b8866

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A rectangular poster has an area of **360** square inches. A copy of the poster is made in which the length and width of the original poster are each increased by **20%**. What is the area of the copy, in square inches?

Question ID: 2855cb58

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has its center at $(16, 17)$ and has a radius of $7k$. Which equation represents this circle?

Answer

- A. $(x - 16)^2 + (y - 17)^2 = 49k$
- B. $(x - 16)^2 + (y - 17)^2 = 49k^2$
- C. $(x - 16)^2 + (y - 17)^2 = 7k$
- D. $(x - 16)^2 + (y - 17)^2 = 7k^2$

Question ID: dc71597b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a volume of $\frac{1}{3}\pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

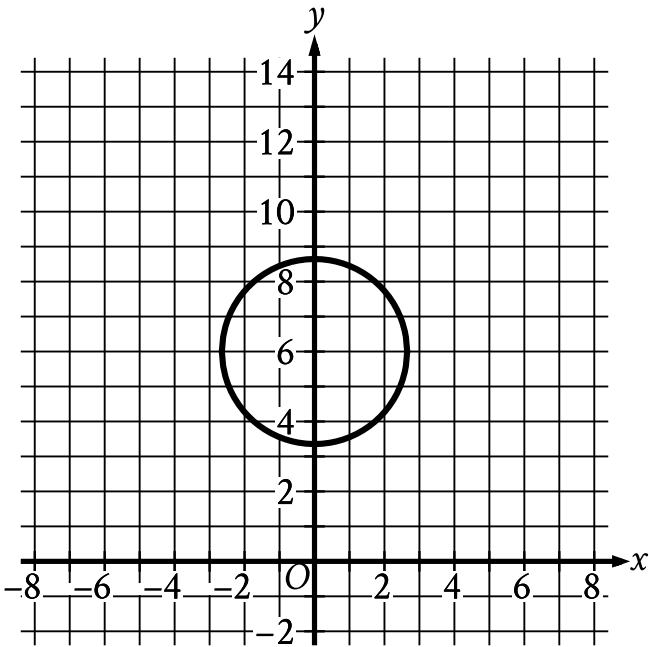
Answer

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. 3

Question ID: 1b2b20b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Circle A shown is defined by the equation $x^2 + (y - 6)^2 = 7$. Circle B (not shown) has the same radius but is translated **96** units to the right. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, where h , k , and a are constants, what is the value of **4a**?

Question ID: 7ce2e728

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

For triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH ?

- I. The measure of angle A is equal to the measure of angle F .
- II. The measure of angle C is equal to the measure of angle H .
- III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.

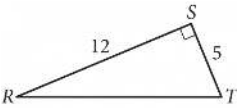
Answer

- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

Question ID: 6933b3d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of $\cos(\angle RSW) - \sin(\angle WST)$?

Question ID: 6ab30ce3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Triangle ABC is similar to triangle DEF , where A corresponds to D and C corresponds to F . Angles C and F are right angles. If $\tan(A) = \sqrt{3}$ and $DF = 125$, what is the length of $\overline{DE}$?

Answer

- A. $125\frac{\sqrt{3}}{3}$
- B. $125\frac{\sqrt{3}}{2}$
- C. $125\sqrt{3}$
- D. 250

Question ID: a5ff1a96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$. Circle P has the same center as circle M but has a radius that is twice the radius of circle M . Which equation represents circle P ?

Answer

- A. $(x - 4)^2 + (y - 5)^2 = 18$
- B. $(x - 4)^2 + (y - 5)^2 = 36$
- C. $(x - 8)^2 + (y - 10)^2 = 9$
- D. $(x - 8)^2 + (y - 10)^2 = 18$

Question ID: 73e1b793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is **972 cubic centimeters (cm^3)**, and the volume of prism Q is **36 cm^3** . One edge of prism P has a length of **12 cm**. What is the length, in **cm**, of the corresponding edge of prism Q?

Question ID: 5b4757df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

Question ID: ca2235f6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

Answer

- A. **3**
- B. **6**
- C. **9**
- D. **24**

Question ID: 9fdc4eaa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. What is the height, in inches, of this pyramid?

Answer

- A. **56**
- B. **90**
- C. **106**
- D. **180**

Question ID: f6ca90cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right prism has a square base with side lengths of **10** inches. The height of the prism is **58** inches. What is the volume, in cubic inches, of the prism?

Question ID: 981275d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$(x - 6)^2 + (y + 5)^2 = 16$

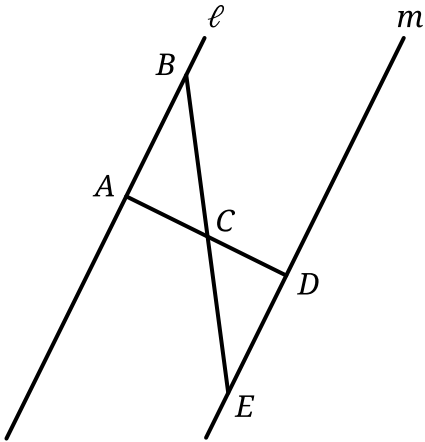
In the xy-plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

Answer

- A. $(2, -5)$
- B. $(6, -1)$
- C. $(6, -5)$
- D. $(6, -9)$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, segment AD and segment BE intersect at point C . Segment AD is perpendicular to line ℓ at point A . Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC ?

- I. Segment AD is perpendicular to line m at point D .
- II. $AC = 13$ and $DC = 13$

Answer

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

Question ID: 89661424

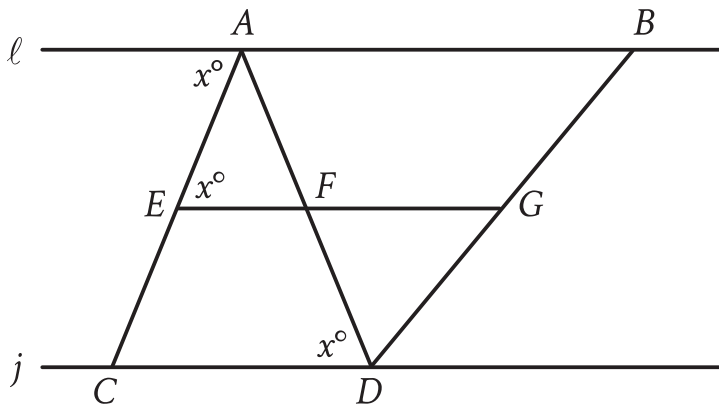
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has its center at $(-5, 2)$ and has a radius of 9 . An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F , and line **Math output error** is parallel to line j . If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

- I. $AB = 2.25EF$
- II. $FG = 0.5AB$
- III. $\frac{EF}{FG} = \frac{AF}{FD}$

Answer

- A. I only
- B. II only
- C. I and II
- D. II and III

Question ID: 7c25b0dc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

The length of a rectangle’s diagonal is $3\sqrt{17}$, and the length of the rectangle’s shorter side is 3 . What is the length of the rectangle’s longer side?

Question ID: 4fb972f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS ?

Question ID: 93de3f84

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

Answer

- A. 11
- B. 22
- C. 44
- D. 66

Question ID: 149021da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center P , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

Question ID: fb58c0db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Points A and B lie on a circle with radius 1, and arc $\overline{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\overline{AB}$?

Question ID: c6dff223

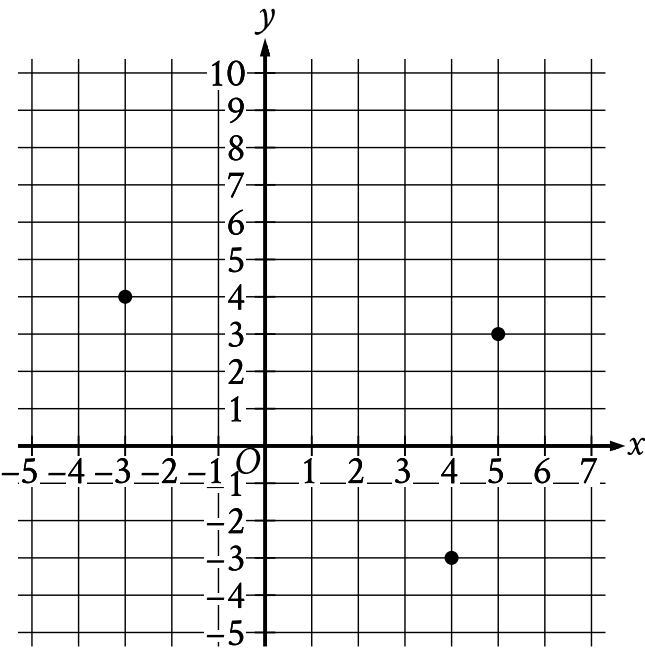
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length of $\overline{AB}$ is 2.9 times the length of $\overline{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



What is the area, in square units, of the triangle formed by connecting the three points shown?

Question ID: 4757123b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Question ID: 92eb236a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

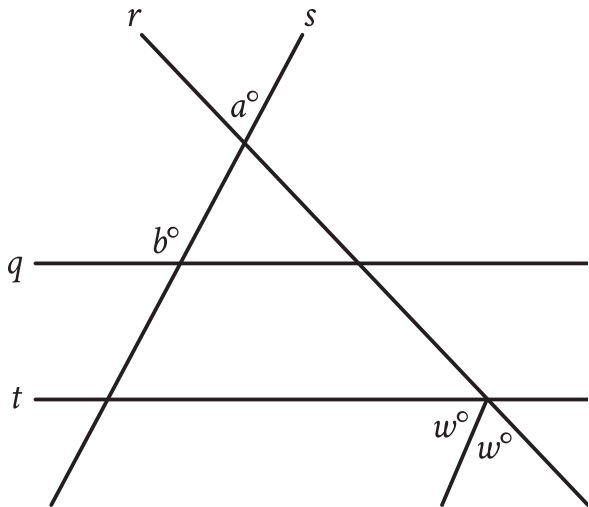
In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

Answer

- A. $-\frac{\sqrt{3}}{3}$
- B. $-\frac{3}{\sqrt{3}}$
- C. $\frac{\sqrt{3}}{3}$
- D. $\frac{3}{\sqrt{3}}$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



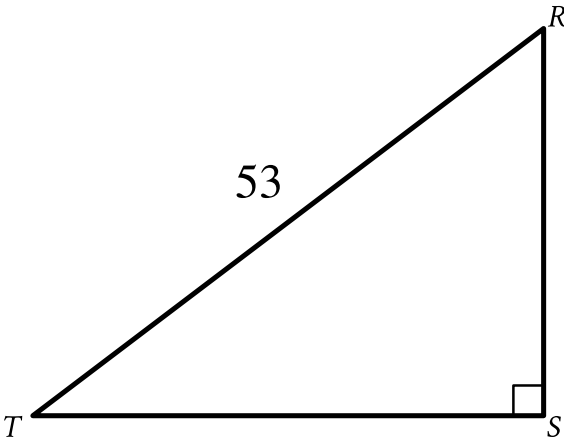
Note: Figure not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

Question ID: a67b9f88

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the triangle shown, $RS = \sqrt{105}$. What is the value of $\sin R$?

Question ID: f7e626b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question
The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

- Answer**
- A. 30
 - B. 74
 - C. 120
 - D. 148

Question ID: 2be01bd9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

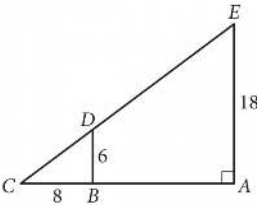
Question

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angle C corresponds to angle F . Angles C and F are right angles. If $\tan(A) = \frac{50}{7}$, what is the value of $\tan(E)$?

Question ID: dba6a25a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the figure above, $\overline{BD}$ is parallel to $\overline{AE}$. What is the length of $\overline{CE}$?

Question ID: acd30391

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

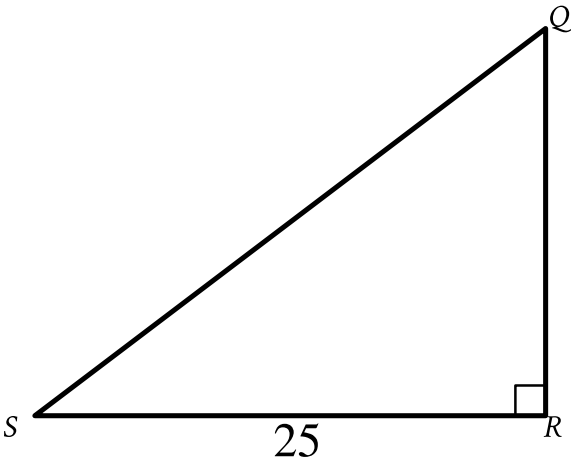
A circle in the xy -plane has equation $(x + 3)^2 + (y - 1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

Answer

- A. $(-7, 3)$
- B. $(-3, 1)$
- C. $(0, 0)$
- D. $(3, 2)$

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

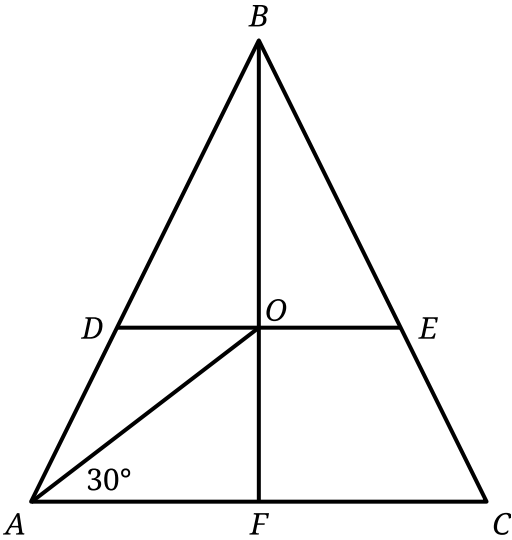
In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- Answer
- A. $25 \cos Q$
 - B. $25 \sin Q$
 - C. $\frac{25}{\cos Q}$
 - D. $\frac{25}{\sin Q}$

Question ID: 1dacfb94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In triangle ABC , $AB = BC$. Points D and E lie on line segments AB and BC , respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE , and the measure of $\angle OAC$ is 30° . Point F lies on line segment AC , and line segment BF passes through point O . If $AB = 81$ and $AO = BO$, what is the length of line segment BE ?

Question ID: 10e059a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle A is $\frac{\pi}{216}$ radians. The measure of angle B is 36 times the measure of angle A . What is the value of $\cos B$?

Answer

- A. 0
- B. $\frac{1}{2}$
- C. $\frac{\sqrt{2}}{2}$
- D. $\frac{\sqrt{3}}{2}$

Question ID: 9f9112ab

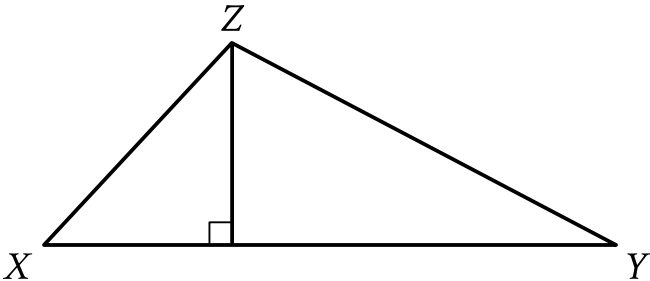
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle L is 60 degrees. If the measure of angle L is $\frac{\pi}{n}$ radians, where n is a constant, what is the value of n ?

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

Answer

- A. 247
- B. 494
- C. $247 \sin 54^\circ$
- D. $494 \sin 54^\circ$

Question ID: 27f1db42

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$. Circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = p$, where p is a constant. What is the value of p ?

Question ID: 14e7c1f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

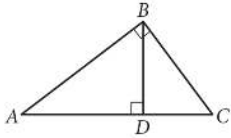
For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x ?

- Answer**
- A. 5
 - B. 19
 - C. 23
 - D. 29

Question ID: 6a3fbec3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$. What is the length of $\overline{DC}$?

Question ID: c0b53183

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center O , and points A and U lie on the circle. Line segments AE and UE are tangent to the circle at points A and U , respectively. The distance between point O and point E is **510** centimeters and the distance between point O and point U is **78** centimeters. What is the area, in square centimeters, of quadrilateral $OAEU$?

Question ID: 459dd6c5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF ?

Question ID: 44d67c6c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the xy -plane, where n is a constant. Point $(11, 8)$ lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n ?

Question ID: 25da87f8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18+6\sqrt{3}$. What is the length of the longest side of the triangle?

Question ID: 310c87fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

Answer

- A. 18
- B. 27
- C. 36
- D. 81